Physical Chemistry Cume

December 2, 2005

Regarding the Electric Properties of Molecules

1.) Define the electric **dipole moment** of a molecule.

2.) Define the **polarization** of a dielectric.

3.) Define the **polarizability** of a molecule.

4.) The **molar polarization**, P_m, is given by the Debye equation:

$$P_m = (N_A/3\varepsilon_o)\,(\alpha + \mu^2/[3kT])$$

where α is the polarizability of the molecule, N_A is Avogadro's number, k is Boltzmann's constant, T is temperature in K, and μ is the permanent electric dipole moment of the molecule. For camphor, molecular weight M=152.23 g mol^{-1}, measurements yielded P_m= 118 cm^3/mol at 40°C, and P_m = 111 cm^3/mol at 100°C. Calculate α and μ.

Constants: k=1.38E-23 JK^{-1}; ε_o= 8.854 E-12 $J^{-1}C^2m^{-1}$; N_A= 6.022 E23 mol^{-1}. For μ use the coulomb-metres unit, C m. When field strength is expressed in terms of volts per metre (Vm^{-1}), α has the units $C^2m^2J^{-1}$.

(Points: Problems 1, 2, and 3 count a total of 40 points; problem 4, sixty points.)